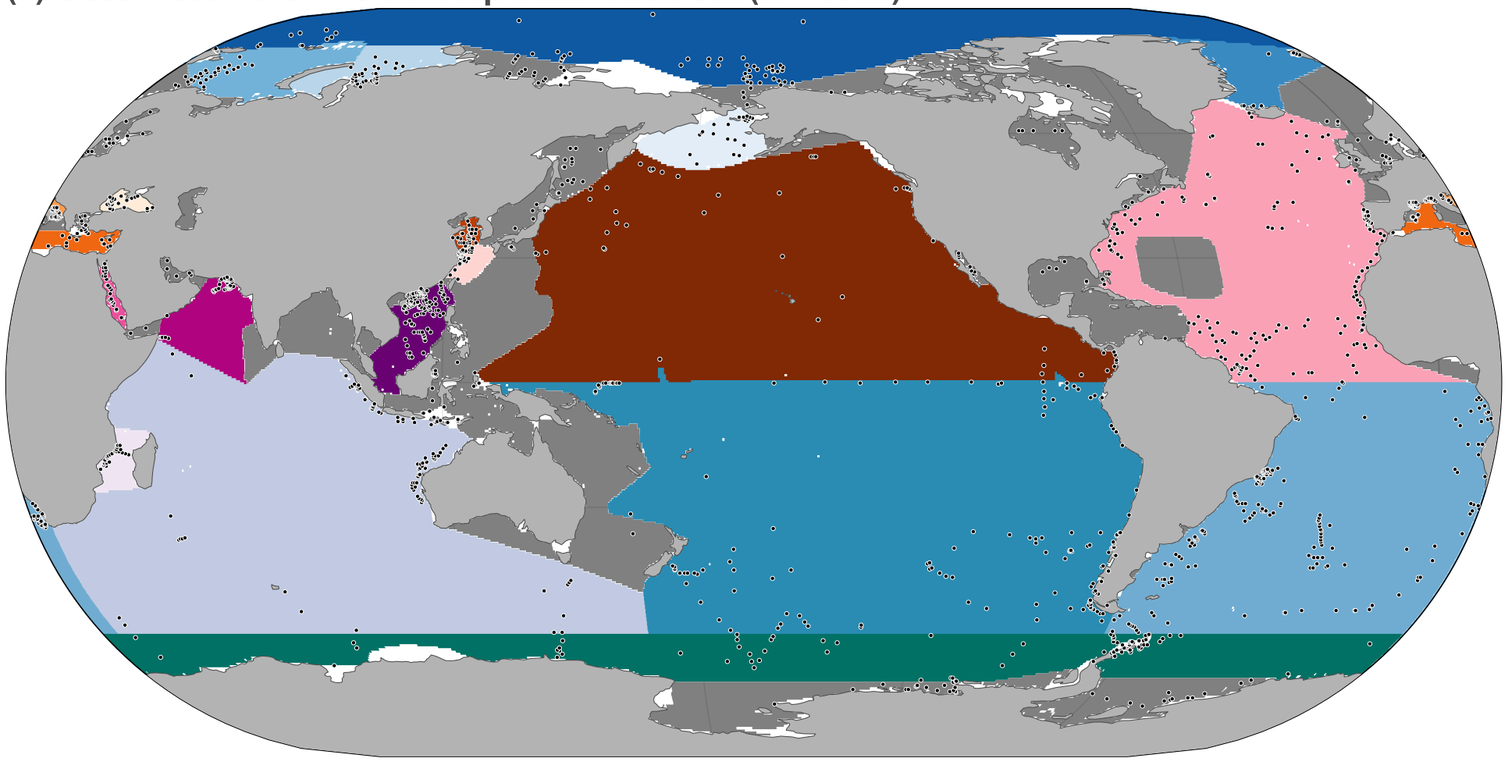
(a) Ocean basins of the coretop calibration set (n = 1518)



Arctic (n = 58)
Barents Sea (n = 28)
Bering Sea (n = 24)
Golfe du Lion (n = 23)
Mediterranean (n = 31)
N. Pacific (n = 46)
N. Atlantic (n = 173)
Arabian Sea (n = 41)
Mozambique Ch. (n = 30)
S. Atlantic (n = 207)
Southern (n = 77)
Greenland Sea (n = 20)
Kara Sea (n = 38)
Black Sea (n = 25)
Adriatic Sea (n = 24)
Yellow Sea (n = 31)
E. China Sea (n = 34)
Red Sea (n = 23)
S. China Sea (n = 117)
Indian (n = 95)
S. Pacific (n = 141)
basins with n < 20 (n = 232)

(b) SST prediction RMSE by ocean basin (n ≥ 20)

21 basins hold 1286 of 1518 sites (85%); black box = lowest RMSE in that basin

	Arctic	Greenland Sea	Barents Sea	Kara Sea	Bering Sea	Black Sea	Golfe du Lion	Adriatic Sea	Mediterranean	Yellow Sea	N. Pacific	E. China Sea	N. Atlantic	Red Sea	Arabian Sea	S. China Sea	Mozambique Ch.	Indian	S. Atlantic	S. Pacific	Southern	All RMSE	Std of region-RMSE
Linear (Schouten+ 2002)	9.5	6.6	4.3	8.6	5.6	5.2	5.9	5.8	5.2	5.8	3.7	3.1	4.4	9.5	4.3	5.1	2.1	3.5	3.9	2.2	7.1	5.3	2.1
Exponential (Kim+ 2010)	12.5	9.8	6.9	11.7	8.7	2.2	3.2	3.5	5.6	3.2	4.9	1.7	3.8	5.7	2.9	3.3	1.3	3.7	3.5	3.5	9.9	5.9	3.3
BAYSPAR (Tierney+ 2014)	8.7	7.7	4.8	8.9	4.2	4.5	5.1	4.9	5.3	2.8	2.6	1.6	3.7	1.4	3.6	3.2	1.7	3.3	2.8	2.2	3.2	4.2	2.1
TEXAS (T only)	3.7	4.2	3.2	2.5	3.3	3.2	3.9	2.5	2.7	3.3	3.8	5.4	4.6	6.7	5.2	2.7	2.0	3.5	4.9	3.8	3.0	3.9	1.1
TEXAS (T + G23)	4.6	5.2	5.1	3.6	5.1	3.7	5.3	3.3	4.4	4.0	3.3	4.4	4.3	5.2	3.3	2.5	0.9	3.2	5.2	4.5	4.2	4.2	1.1
TEXAS (T + G23 + NO_3^-)	5.0	5.7	5.3	4.1	5.4	4.3	5.0	3.3	3.0	3.9	3.6	5.4	4.2	2.1	2.5	2.8	2.2	3.5	5.4	4.8	4.7	4.3	1.1

RMSE (°C): 13.5, 12, 10.5, 9, 7.5, 6, 4.5, 3, 1.5, 0